João André **Roque Costa**

Software, Data & AI Engineer
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Computer Science and Engineering graduate focused on **software development**, artificial intelligence, data science, machine learning, optimization, and scalable computation. Brings **Python** software and data-engineering experience, academic C/C++ systems work, algorithms and data-structures foundations, **Linux and Git**, and research on nonlinear dimensionality reduction for large datasets.

Professional Experience

Tech Analyst — *Coimbra, Portugal*
Deloitte — *Apr. 2026 - Oct. 2026*

- Built a **Python** prototype for data migration and synthetic-data generation with **Azure AI Foundry**, Azure Blob Storage, and MongoDB.
- Automated reporting and data-extraction workflows with Python and Playwright, and contributed technical context to **RAG** and **OCR** designs for document analysis.
- Delivered an **Azure** subscription migration for Databricks and Azure Data Factory assets, coordinating dependent services across a data platform after rapidly ramping up on Azure.

Teaching Assistant — *Lisbon, Portugal*
Instituto Superior Técnico — *Oct. 2024 - Jul. 2025*

- Taught and evaluated practical classes in Databases and Data Science, adapting explanations to different audiences and tracking student progress.
- Guided **Python** work in data preparation, statistical analysis, and visualization with pandas and matplotlib, communicating technical results clearly and constructively.

Education

M.Sc. Computer Science and Engineering — *Lisbon, Portugal*
Instituto Superior Técnico — *Sep 2023 - Nov 2025*

- Master of Science in Computer Science and Engineering, focused on **Artificial Intelligence** and Data Science, including machine learning, cloud computing, multi-agent systems, and large-scale data analysis.
- Relevant Coursework: **Optimization Algorithms**, Machine Learning, Deep Learning, Data Analysis and Integration, Data and Information Systems Management, Cloud Computing and Virtualization, Autonomous Agents and Multi-Agent Systems.
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)
- Thesis: Developed a decomposition-based extension to MVU for **nonlinear dimensionality reduction** on disconnected datasets.
- Thesis: Designed a parallelizable method for large datasets, preserving local geometric structure while producing comparable embeddings.

B.Sc. Computer Science and Engineering — *Lisbon, Portugal*
Instituto Superior Técnico — *Sep 2020 - Jul 2023*

- Relevant Areas: Algorithms, distributed systems, operating systems, database theory, C/C++, and SQL.
- Relevant Project: Applied **C/C++** in systems projects and designed relational schemas and transactions in **SQL**.

Extracurricular

- Built a self-hosted **Python** monitoring service with resource management, historical trend analysis, alerts, and an operational dashboard for tracking system state in a **Linux** environment.
- Developed an event-driven Discord matchmaking service using ELO ratings, **PostgreSQL** match history, and **Redis** queues and caching.
- Built automation and API integrations for multi-platform price monitoring, risk management, optimal stake sizing, and inventory tracking.

Skills & Competencies

Programming and Systems	Python, C, C++, SQL, Linux, Bash scripting, Git, RESTful APIs, data structures, algorithms.
Artificial Intelligence and Data	machine learning, deep learning, statistical analysis, data analysis, numerical computing, NumPy, pandas, TensorFlow, Keras, scikit-learn.
Optimization and Research	optimization algorithms, numerical optimization, nonlinear dimensionality reduction, large-scale datasets, scalable computation.
Cloud and Automation	Azure AI Foundry, Databricks, Azure Data Factory, Azure Blob Storage, Playwright, RAG, OCR.
Databases and Infrastructure	PostgreSQL, MySQL, MongoDB, Redis, Docker.
Communication	technical writing, technical presentations, teaching, mentoring, cross-functional communication, willingness to learn.